Secure By Design

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COURSE 1.19 / COMPANION WORKBOOK

Prove you can recover

Backups with custody, one rehearsed restore & the closing record

Close the Build path by proving the network can come back: every backup in the network defined and scheduled in one place — the directory, the CA under ceremony-grade custody, Group Policy, DHCP, the repository and analysis configuration, the VLAN 40 business services with their permissions, logins and certificates, the log archive, and the living records — in one hashed catalog, a timed restore rehearsal in isolation, and the recovery record that completes the Body of Evidence.

MATERIALS EDITION 1.4 / COURSE 1.19

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Read each lesson alongside the interactive guide and the video. Use the record tables to write down what you actually observe — the workbook is your evidence, not the guide.

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Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

Before you start

The model's final chapter: a network that cannot come back was never really built. This course is the one place every backup in the network is defined and scheduled — the directory, both CAs, Group Policy, DHCP, the repository and analysis configuration, the VLAN 40 business services with their permissions, logins and certificates, the log record, and the living records — then proves one real restore in isolation and files the tested-recovery evidence the authorization path expects.

Prerequisites

- ☐ Complete courses 1.1–1.18 (courses 1.16 and 1.17 are coming soon and aren't needed): the full network, the VLAN 40 business services, their records, the protected log archive, and the living inventory all exist.
- ☐ Provision the backup storage this course schedules into: a protected backup volume (E: in this sample) on every Windows server in the recovery set — both domain controllers, SBD-SUBCA01, SBD-DHCP01, SBD-MGMT01, SBD-FS01, SBD-SQL01 and SBD-APPS01 — never one of the volumes being backed up and written only by the backup jobs, and a separate volume mounted at /srv/backup, readable by root alone, on SBD-REPO01, SBD-SPLUNK01 and SBD-WEB01. Earlier courses set up no backup jobs; this course defines and schedules every one. Also provide an offline backup cabinet with removable media and a custody register, and an isolated recovery network segment (or host) for the restore rehearsal.
- ☐ Have the custody records handy: the sealed Directory Services Restore Mode (DSRM) passwords set in course 1.1 and the course 1.6 ceremony record for SBD-ROOTCA01's protected backup — recovery is where those promises come due.

What this course builds

- Define and schedule every backup in the network — directory, CA, policy, DHCP, repository, analysis configuration, business services, logs, and records — in one verified catalog.
- Rehearse a real system state restore in isolation, measured against written objectives.
- File the tested-recovery record that completes the Body of Evidence for path 2.

Systems in scope

RECORD	YOUR OBSERVATION
SBD-DC01 — name, address, OS build	
SBD-DC02 — name, address, OS build	
SBD-SUBCA01 — name, address, OS build	
SBD-DHCP01 — name, address, OS build	

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RECORD	YOUR OBSERVATION
SBD-AUDIT01 — name, address, OS build	
SBD-REPO01 — name, address, OS build	
SBD-SPLUNK01 — name, address, OS build	
SBD-MGMT01 — name, address, OS build	
SBD-FS01 — name, address, OS build	
SBD-SQL01 — name, address, OS build	
SBD-APPS01 — name, address, OS build	
SBD-WEB01 — name, address, OS build	

LESSON 01 / ABOUT 25 MIN

Decide what recovery must prove

> **WHY THIS MATTERS** Recovery starts as a list of promises: which systems and records must be restorable, how old a restored copy may be, how long restoration may take, and the one schedule that says how each is backed up — written as commitments the rest of the course exists to keep.

In this lesson you will

- Name the recovery set: what must come back, in what order.
- Set the recovery point and time objectives per item.
- Write the backup schedule: every backup in the network, defined in one place.
- Decide where backups live and who can touch them.

Follow the build

1. Write the recovery set in dependency order — the order the path built it: the directory on both controllers first, because everything trusts it; then the certificate authorities (SBD-SUBCA01's database and key, and SBD-ROOTCA01's course 1.6 ceremony backup); Group Policy; SBD-DHCP01 (course 1.7); the boundary's records (the service-path register and the core router's configuration); SBD-REPO01's snapshot catalog and snapshots (course 1.9); SBD-SPLUNK01's configuration (course 1.12); the log archive and its catalog on SBD-AUDIT01 (course 1.11); the inventory on SBD-MGMT01 (course 1.15); the VLAN 40 business services (course 1.18) — SBD-FS01's share data with its NTFS permissions, SBD-SQL01's databases and the logins they depend on, SBD-APPS01's and SBD-WEB01's configuration and certificates; and the exception record.
2. Set the two numbers per item: how much history may be lost (recovery point) and how long restoration may take (recovery time) — lab-honest values you will actually measure against in lesson 5, not aspirations. A directory backup older than the forest's tombstone lifetime (180 days by default) can't be restored at all, so the directory's recovery point stays well inside it.
3. Write the backup schedule, the one place every backup in this network is defined: for each item, what is captured, the tool, when it runs, where it lands, and who restores it. No earlier course scheduled a backup; the one build-time copy is SBD-ROOTCA01's, taken inside the course 1.6 ceremony, and the schedule accounts for it rather than repeating it. Lessons 2 and 3 put each row into effect.
4. Decide backup placement: each protected backup volume is written only by its backup jobs and readable by the recovery role, and custodians of the matching tier carry copies to the offline backup cabinet on media — the air-gapped network's version of an off-site copy. There is no backup server and no network path to the cabinet.
5. Confirm the standing custody items: the DSRM passwords set in course 1.1 are sealed and findable, and SBD-ROOTCA01's protected backup from the course 1.6 ceremony is accounted for — recovery depends on secrets that must exist before the bad day.

Check your result

```
# On each Windows server in the recovery set: capture the current protection state before this course changes it.
wbadmin get versions
Get-Volume -DriveLetter E | Select-Object DriveLetter, FileSystemLabel, SizeRemaining   # the protected backup volume
```

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```
# The tombstone lifetime bounds the directory's recovery point (empty means the 180-day
default).
(Get-ADObject "CN=Directory Service,CN=Windows NT,CN=Services,$((Get-
ADRootDSE).configurationNamingContext)" -Properties tombstoneLifetime).tombstoneLifetime
```

WHAT SUCCESS LOOKS LIKE The recovery set is ordered by dependency with point and time objectives per item, the backup schedule defines every backup in one place, the placement and access rules are written, and the DSRM and root-key custody records check out.

Your record

Evidence to keep: Keep the recovery requirements sheet, the objectives table, the backup schedule, the placement and access decisions, and the custody verification notes.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why does the directory lead the recovery order?

REMEMBER The recovery set is ordered by dependency, directory first, VLAN 40 included. Point and time objectives exist to be measured, not admired. One schedule defines every backup in the network. Backups get a protected home plus one offline, custody-tracked copy.

Microsoft · Active Directory forest recovery guide

NIST · SP 800-53 Rev. 5 security controls (CP family)

LESSON 02 / ABOUT 35 MIN

Back up Tier 0: directory, CA, and policy

> **WHY THIS MATTERS** The Tier 0 rows of the schedule: system state from both controllers, the issuing CA under ceremony-grade protection, and the whole Group Policy estate with its links — plus the quiet feature that handles the most common 'disaster' of all: the Recycle Bin for deleted directory objects.

In this lesson you will

- Schedule system state and critical-volume backups on both domain controllers, and verify and catalog them.
- Back up the issuing CA's database, key, and configuration under ceremony-grade protection.
- Schedule the Group Policy backup with its link export.
- Enable the Active Directory Recycle Bin for object-level recovery.

Follow the build

1. In a Tier 0 session from AdminJumpBox01, schedule Windows Server Backup on SBD-DC01 and SBD-DC02, each to its own protected backup volume, with the system state and all critical volumes (wbadmin's -systemState and -allCritical): the system state is what a directory restore in DSRM needs, and the critical volumes let the same backup rebuild a controller whose hardware is gone. Two controllers, two independent backups — replication is availability and was never recovery, as course 1.1 put it.
2. After the first run, verify each backup's completion, record its version identifier in the backup catalog (C:\Evidence\recovery on AdminJumpBox01), and hash the catalog — an unverified backup is a hope with a timestamp.
3. Back up the issuing CA. Schedule SBD-SUBCA01's own system state and critical-volume backup to its protected volume, so the server can be rebuilt; then take the CA's own backup — database and private key — with certutil to E:\Backups\CA, with its registry configuration exported beside it, on the cadence the schedule sets and after every CA key renewal. It prompts for the password that protects the key, so it is a Tier 0 task, never an unattended one. Never write it to C:\Evidence: the private key is a secret, and secrets never go into evidence. Record custody and put only the hashes in evidence; SBD-ROOTCA01's ceremony backup is verified against its record, never duplicated or brought onto the network.
4. Schedule the Group Policy backup: a daily task on SBD-DC01, running as SYSTEM from a script folder only Tier 0 administrators can write, that runs Backup-GPO -All into a dated folder on E:\Backups\GPO and exports every link beside it — a restored GPO brings back its settings and permissions, not its links. That covers the whole estate, course 1.6's SBD Certificate Autoenrollment and course 1.8's tier logon GPOs included; the GPO reports and link exports courses 1.5 and 1.6 filed stay as their change records.
5. Enable the Active Directory Recycle Bin from the Tier 0 seat — a one-way, forest-level change run against the domain naming master (SBD-DC01) and recorded as the change it is — so a deleted user or OU comes back intact without touching system state. Objects deleted before it was enabled can come back only from a backup.
6. Write the restore decision tree: deleted object → Recycle Bin; corrupted DC on its own hardware → system state restore in DSRM (lesson 5's rehearsal); DC whose hardware is gone → full server (bare-metal) recovery from the critical-volumes backup; a damaged GPO → Restore-GPO from the dated set,

then its links from the export; whole-forest loss → the Microsoft forest recovery guide, referenced and staged with the evidence rather than improvised on the day.

Check your result

```
# Tier 0 session from AdminJumpBox01: daily schedules on each controller and the issuing
CA (E: is never a volume being backed up)
Invoke-Command SBD-DC01.sbd.test, SBD-DC02.sbd.test, SBD-SUBCA01.sbd.test { wbadmin enable
backup -addtarget:E: -schedule:21:00 -systemState -allCritical -quiet }
Invoke-Command SBD-DC01.sbd.test, SBD-DC02.sbd.test { wbadmin get versions }   # after the
first run
# On SBD-SUBCA01 (Tier 0 session): the CA's own backup, to its protected volume, never to
the evidence folder
certutil -backup E:\Backups\CA   # prompts for the password that protects the exported key
reg export HKLM\SYSTEM\CurrentControlSet\Services\CertSvc\Configuration E:\Backups\CA\
certsvc-config.reg
Get-ChildItem E:\Backups\CA -Recurse -File | Get-FileHash -Algorithm SHA256 | Export-Csv
E:\Backups\ca-backup-hashes.csv -NoTypeInformation   # only this list goes to C:\Evidence\
recovery
# On SBD-DC01: C:\Scripts\Backup-SbdGpo.ps1 (writable by Tier 0 administrators only) runs
these lines, then the task runs it daily
#   $d = 'E:\Backups\GPO\{0:yyyy-MM-dd}' -f (Get-Date); New-Item -ItemType Directory $d -
Force | Out-Null; Backup-GPO -All -Path $d
#   @('DC=sbd,DC=test') + (Get-ADOrganizationalUnit -Filter *).DistinguishedName |
ForEach-Object { (Get-GPInheritance -Target $_).GpoLinks } | Select-Object Target, Order,
DisplayName, Enabled, Enforced | Export-Csv "$d\gpo-links.csv" -NoTypeInformation
Register-ScheduledTask 'SBD GPO backup' -Action (New-ScheduledTaskAction -Execute
powershell.exe -Argument '-NoProfile -File C:\Scripts\Backup-SbdGpo.ps1') -Trigger (New-
ScheduledTaskTrigger -Daily -At 22:00) -User 'NT AUTHORITY\SYSTEM' -RunLevel Highest
# Back on AdminJumpBox01: the Recycle Bin, then the catalog
Enable-ADOptionalFeature 'Recycle Bin Feature' -Scope ForestOrConfigurationSet -Target
sbd.test -Server SBD-DC01.sbd.test
Get-ADOptionalFeature -Filter {Name -like 'Recycle Bin*'} | Select-Object Name,
EnabledScopes
Get-FileHash -Algorithm SHA256 C:\Evidence\recovery\backup-catalog.txt
```

> **WHAT SUCCESS LOOKS LIKE** Both DCs and SBD-SUBCA01 show scheduled, completed system state and critical-volume versions on their protected volumes; the CA backup and its configuration sit on SBD-SUBCA01's protected volume with key custody recorded and only hashes in evidence; the GPO task has written a dated set with its link export; the Recycle Bin reports an enabled scope; and the decision tree names the tool for each failure class.

Your record

Evidence to keep: Keep the backup schedules and version listings, the CA backup's hash list and custody record (never the backup itself), the GPO task definition and its first set's hashes, the hashed catalog, the Recycle Bin enablement record, and the restore decision tree.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Both DCs replicate — why back up both?

> **REMEMBER** Two independent system state schedules, verified and cataloged. CA database, key, and configuration: backed up to the protected volume, custody recorded, only hashes in evidence. Every GPO is backed up daily with its links beside it. The Recycle Bin handles the everyday disaster in seconds; a decision tree matches each failure class to its tool.

Microsoft · Active Directory forest recovery guide

Microsoft · Wbadmin (Windows Server Backup)

Microsoft · wbadmin enable backup

Microsoft · AD forest recovery: perform a full server recovery

Microsoft · Enable-ADOptionalFeature (Active Directory Recycle Bin)

Microsoft · Certutil reference (CA backup and restore)

Microsoft · Backup-GPO

Microsoft · Register-ScheduledTask

LESSON 03 / ABOUT 40 MIN

Back up the member servers and records

WHY THIS MATTERS The rest of the schedule: SBD-DHCP01 and the inventory, SBD-REPO01 and SBD-SPLUNK01, and the business services — SBD-FS01's shares with their permissions, SBD-SQL01's databases with the logins they depend on, SBD-APPS01's and SBD-WEB01's configuration and certificates — then the log archive and the boundary's records, all in one hashed catalog.

In this lesson you will

- Schedule the Tier 1 infrastructure backups — SBD-DHCP01, SBD-MGMT01, SBD-REPO01, SBD-SPLUNK01 — with the restore notes each needs.
- Schedule the VLAN 40 backups so permissions, logins and certificates come back with the data.
- Fold the log archive, the boundary records and the living records into the same catalog.

Follow the build

1. In Tier 1 sessions from SBD-ADMIN-SRV01, schedule SBD-DHCP01: a daily task, running as SYSTEM from a script folder only SBD Tier 1 Admins can write, takes Backup-DhcpServer into the default backup folder, copies it to a dated folder on E:\Backups\DHCP with an Export-DhcpServer of the configuration and leases, and hashes both. File course 1.7's restore note with it: after Restore-DhcpServer, set the DNS update credential again with Set-DhcpServerDnsCredential from the sealed record, because the backup doesn't carry it. On SBD-MGMT01, schedule Windows Server Backup of D:\Inventory (course 1.15) to its E:.
2. Schedule SBD-FS01: Windows Server Backup of D:\Shares to its E: every night. It keeps each file's NTFS permissions, and a file recovery restores them by default — never run it with -notRestoreAcl. Beside it, after every access change, save the readable record: icacls /save of the share tree's ACLs and an export of the share definitions (the LanmanServer Shares registry key, which also holds the share permissions), so permissions can be checked or reapplied without restoring data.
3. Schedule SBD-SQL01 inside the engine: a nightly SQL Server Agent job backs up each application database and the master, model and msdb system databases WITH CHECKSUM to E:\Backups\SQL, then checks each file with RESTORE VERIFYONLY. The engine writes the files as its gMSA, so grant SBD\SBD-SQL$ modify on that folder alone. master is where the logins live, msdb holds the Agent jobs, and the application databases' users map to the SBD\SBD-Apps$ login by its directory SID — so record the logins and each database's user mappings, and write the restore order: the instance and master (or the logins, recreated from the record) before the application databases. Use the simple recovery model unless your lesson 1 recovery point needs log backups, and a rebuilt SBD-SQL01 requests a new certificate through the course 1.6 process.
4. Schedule SBD-APPS01's Windows Server Backup with system state and critical volumes to its E:, so its services' configuration and its certificate, private key included, come back with the server — which is why the course 1.6 template never needs exportable keys. On SBD-WEB01, a root cron job in /etc/cron.d writes a dated tar archive of /etc/httpd, /etc/pki/tls (the certificate and its root-only key) and /etc/sssd, with SELinux contexts and ACLs, to /srv/backup with its SHA-256; SBD-REPO01 does the same for /srv/repo-records and the current snapshot each time it changes, so the previous one is already on media (course 1.9), and SBD-SPLUNK01 for /opt/splunk/etc (course 1.12 — the indexes are a working copy, and the record itself lives on SBD-AUDIT01). The WEB01 and Splunk archives hold keys, so they stay on protected media.

5. Account for the log archive: SBD-AUDIT01's hashed archives in D:\LogArchive and their catalog (D:\LogArchive\Catalog\catalog.csv) are themselves the record, so their backup is the audit custodian's offline copy on the schedule's cadence (lesson 4), and no job runs on that server.
6. Export the boundary and living records into the set: the service-path register, the exception record (C:\Evidence\exception-record.csv), this course's own sheets, and a read-only export of the core router's configuration taken with the router's own export function from AdminJumpBox01 — this course reads the router's configuration and changes nothing on it. Record every first run's hashes in C:\Evidence\recovery on SBD-ADMIN-SRV01. A recovered network without its records would pass no assessment.

Check your result

```
# Tier 1 sessions from SBD-ADMIN-SRV01. On SBD-DHCP01, the task script (C:\Scripts,
writable by SBD Tier 1 Admins only):
$b = 'E:\Backups\DHCP\{0:yyyy-MM-dd}' -f (Get-Date); New-Item -ItemType Directory $b -
Force | Out-Null; $dir = Join-Path $env:windir 'System32\dhcp\backup'
Backup-DhcpServer -Path $dir; Copy-Item $dir (Join-Path $b 'database') -Recurse; Export-
DhcpServer -File (Join-Path $b 'SBD-DHCP01.xml') -Leases
# On SBD-MGMT01, SBD-FS01 and SBD-APPS01: one daily schedule each, to the local E:
wbadmin enable backup -addtarget:E: -include:D:\Inventory -schedule:22:00 -quiet   # SBD-
MGMT01
wbadmin enable backup -addtarget:E: -include:D:\Shares -schedule:22:00 -quiet   # SBD-
FS01; then, after each access change:
icacls D:\Shares\* /save E:\Backups\FS01\shares-acl.txt /t /c; reg export HKLM\SYSTEM\
CurrentControlSet\Services\LanmanServer\Shares E:\Backups\FS01\shares.reg /y
wbadmin enable backup -addtarget:E: -schedule:22:30 -systemState -allCritical -quiet   #
SBD-APPS01
# On SBD-SQL01: the engine's identity may write the backup folder; the Agent job's step
runs this T-SQL for each database
icacls E:\Backups\SQL /grant 'SBD\SBD-SQL$:(OI)(CI)M'
#   BACKUP DATABASE [master] TO DISK = N'E:\Backups\SQL\master.bak' WITH CHECKSUM, INIT;
-- then model, msdb, each application database
#   RESTORE VERIFYONLY FROM DISK = N'E:\Backups\SQL\master.bak' WITH CHECKSUM;
sqlcmd -S SBD-SQL01.sbd.test -E -Q "SELECT name, type_desc, sid FROM sys.server_principals
WHERE type IN ('U','G','S')"   # the login record
# On SBD-WEB01 (and the same pattern on SBD-REPO01 and SBD-SPLUNK01): the cron job's work,
run once by hand
sudo tar --selinux --acls --xattrs -czf /srv/backup/web01-$(date +%F).tar.gz /etc/httpd
/etc/pki/tls /etc/sssd
sudo sha256sum /srv/backup/web01-*.tar.gz
```

> **WHAT SUCCESS LOOKS LIKE** Every member-server row in the schedule has run once and verified: SBD-DHCP01's set with its restore note, D:\Inventory, SBD-FS01's shares with the ACL record and share definitions, SBD-SQL01's application and system databases verified with RESTORE VERIFYONLY and the login record filed, SBD-APPS01's full-server backup, and the dated archives on SBD-WEB01, SBD-REPO01 and SBD-SPLUNK01 — and the catalog covers them with the log archive, router export, and living records.

Your record

Evidence to keep: Keep the task, job and cron definitions, each first run's hashes, the DHCP and SQL restore notes, the ACL record and share export hashes, the login record, and the complete recovery catalog — never the WEB01 or Splunk archives themselves, which hold keys.

RECORD	YOUR OBSERVATION
Host(s) and time collected	

RECORD	YOUR OBSERVATION
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

SBD-SQL01's application database is backed up nightly, but master isn't. What breaks on a rebuilt instance?

> **REMEMBER** SBD-DHCP01, the inventory, the repository and Splunk's configuration each run on schedule, with their restore notes. The business services come back whole: shares with their permissions, databases with their logins, applications with their certificates. The log archive, the boundary's records and the living records join one hashed catalog. Every backup in the network is now defined and scheduled in this course.

Microsoft · wbadmin enable backup

Microsoft · Backup-DhcpServer

Microsoft · Set-DhcpServerDnsCredential

Microsoft · Register-ScheduledTask

Microsoft · wbadmin start recovery

Microsoft · icacls

Microsoft · BACKUP (Transact-SQL)

Microsoft · Back up and restore: system databases (SQL Server)

LESSON 04 / ABOUT 20 MIN

Carry the copies to the offline cabinet

> **WHY THIS MATTERS** A backup that sits on the server it protects fails with it. Move a copy of each backup to the offline cabinet by hand, under the same custody rules as any media crossing a boundary, and keep the custody record that proves it.

In this lesson you will

- Define which custodian carries which backup, by tier.
- Transfer the copies to offline media with hashes and custody records.
- Prove a copy can be found, verified, and returned for a restore.

Follow the build

1. Assign custodians by tier: a Tier 0 custodian handles the controllers' system state, the issuing CA's database and key backup, the Group Policy set, and the router configuration export; the audit custodian (SBD Audit Custodians, course 1.8) handles the log archive and its catalog from SBD-AUDIT01; a Tier 1 server custodian handles the member-server sets — SBD-DHCP01, SBD-REPO01, SBD-SPLUNK01, SBD-MGMT01, and the VLAN 40 servers SBD-FS01, SBD-SQL01, SBD-APPS01 and SBD-WEB01. No one carries a tier they do not hold.
2. On the cadence the backup schedule sets, for each copy, write the media ID, hash the backup on its protected volume, copy it to encrypted removable media, and verify the hash on the media before it leaves the room. Keep Windows Server Backup's WindowsImageBackup folder layout on the media so wbadmin can read it there.
3. Carry the media to the offline backup cabinet and record the handoff in the custody register: who, what, when, the hashes, and the storage slot. This is the course 1.4 media procedure used for backups.
4. Pull one copy back as a test: find it from the register, verify its hash, and confirm it is readable. Lesson 5's rehearsal restores from a copy that has made this round trip.

Check your result

```
# On SBD-DC02 (Tier 0), with encrypted media MEDIA-0012 attached as F:: hash on the
volume, copy, hash on the media
Get-ChildItem E:\WindowsImageBackup\SBD-DC02 -Recurse -File | Get-FileHash -Algorithm
SHA256 | Export-Csv F:\MEDIA-0012-volume.csv -NoTypeInformation
Copy-Item E:\WindowsImageBackup -Destination F:\ -Recurse
Get-ChildItem F:\WindowsImageBackup\SBD-DC02 -Recurse -File | Get-FileHash -Algorithm
SHA256 | Export-Csv F:\MEDIA-0012-media.csv -NoTypeInformation
Compare-Object (Import-Csv F:\MEDIA-0012-volume.csv).Hash (Import-Csv F:\MEDIA-0012-
media.csv).Hash   # no output means the copy matches
wbadmin get versions -backupTarget:F: -machine:SBD-DC02   # the media is readable as a
backup
```

> **WHAT SUCCESS LOOKS LIKE** Every item in the recovery set has an offline copy with matching hashes on the volume and the media, a custody entry with a named custodian of the right tier, and one copy has been retrieved and verified.

Your record

Evidence to keep: Keep the custodian assignments, the hash pairs (copied into C:\Evidence\recovery on the tier's seat), the custody register entries, and the retrieval test record.

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RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Who carries the domain controllers' system-state backup to the cabinet?

REMEMBER Custodians carry only the backups their tier allows. Every copy has matching hashes and a custody entry. A copy has been retrieved and verified before anyone needs it.

Microsoft · Wbadmin (Windows Server Backup)

NIST · SP 800-53 Rev. 5 security controls (CP family)

LESSON 05 / ABOUT 35 MIN

Rehearse a real restore, in isolation

> **WHY THIS MATTERS** One restore, actually performed: a domain controller's system state brought back on the isolated segment, timed against the objective, verified against the directory — plus the small restore everyone will someday need, from the Recycle Bin.

In this lesson you will

- Restore a DC's system state in Directory Services Restore Mode on the isolated recovery segment.
- Measure the restore against the time objective and verify directory health.
- Perform and record an object-level Recycle Bin restore.

Follow the build

1. Stage the rehearsal: the isolated recovery segment has no path to the production VLANs — a restored controller believing it is authoritative must never meet the live directory; that isolation rule is written before the restore starts. Turn the hypervisor's time synchronization off on the rehearsal copy, as for every virtual machine in the path.
2. Restore SBD-DC02 from the offline copy retrieved in lesson 4, timing the work from decision to directory services running against the lesson 1 objective. A system state restore needs the same server instance, so use a copy of SBD-DC02's virtual machine attached only to the isolated segment, start it in DSRM (bcdedit /set safeboot dsrepair, restart, sign in with the DSRM password), and run wbadmin start systemstaterecovery with -authsysvol: the copy is the only controller in its isolated world, so its SYSVOL must be authoritative or SYSVOL and NETLOGON never come up. Restoring to different hardware needs a full server (bare-metal) recovery from the critical-volumes backup instead.
3. Remove the DSRM boot setting (bcdedit /deletevalue safeboot), restart normally, and verify the restored directory the course 1.1 way — services, the SYSVOL and NETLOGON shares, a directory query. Then check the clock: the copy comes back with the time it was saved with, and with no SBD-DC01 to follow nothing corrects it, so compare it with the reference and record the offset; a restored SBD-DC01 in production is checked against the reference before any member follows it (course 1.3). Record what the rehearsal taught — surprises, the measured time, decision-tree updates — then tear the rehearsal down completely.
4. Back in production, from AdminJumpBox01, run the everyday restore: delete a disposable test object, bring it back from the Recycle Bin with attributes intact, and file both rehearsals — the big one and the small one — as the tested-recovery evidence.

Check your result

```
# On the isolated copy of SBD-DC02: start in DSRM, then sign in with the DSRM password.
bcdedit /set safeboot dsrepair
shutdown /r /t 0
wbadmin get versions -backupTarget:F: -machine:SBD-DC02   # F: = the retrieved media
wbadmin start systemstaterecovery -version:<version-id> -backupTarget:F: -machine:SBD-DC02
-authsysvol
bcdedit /deletevalue safeboot   # then restart normally
# After the restart, on the restored copy
Get-Service NTDS, DNS, Netlogon, DFSR | Select-Object Name, Status
Get-SmbShare -Name SYSVOL, NETLOGON
Get-ADDomain -Server localhost | Select-Object DNSRoot, PDCEmulator
```

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```
w32tm /query /status   # then compare Get-Date with the reference clock and record the
offset
# Back in production, from AdminJumpBox01: the Recycle Bin rehearsal
Get-ADObject -Filter "Name -eq '<test object name>'" -IncludeDeletedObjects | Restore-
ADObject
```

WHAT SUCCESS LOOKS LIKE The isolated restore ran in DSRM and reached a healthy directory within (or honestly against) the time objective, its clock offset is recorded, the DSRM boot setting is removed, the teardown is confirmed, and the production object restore returned the test object intact.

Your record

Evidence to keep: Keep the timed rehearsal log with its verification output and clock reading, the lessons-learned and decision-tree updates, the teardown confirmation, and the object-restore record.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why must the rehearsal segment be isolated from production?

REMEMBER One real restore, isolated, timed, verified, and torn down. The measured number replaced the hoped-for one. The everyday restore is proven alongside the dramatic one.

Microsoft · Active Directory forest recovery guide

Microsoft · wbadmin start systemstaterecovery

Microsoft · AD forest recovery: perform a nonauthoritative restore

Microsoft · Restore a virtual domain controller

Microsoft · AD forest recovery: perform a full server recovery

Microsoft · Windows Time service tools and settings

LESSON 06 / ABOUT 20 MIN

File the recovery record and close the path

WHY THIS MATTERS The Build path ends the way it worked: owners, schedules, and evidence. The recovery record joins the living documents, the closing SCAP results snapshot the finished network, and the Body of Evidence is ready for the authorization courses to assemble.

In this lesson you will

- Assemble the tested-recovery record with owners and review dates.
- Take the closing SCAP snapshot across the finished network.
- Hand the complete evidence set to the authorization path.

Follow the build

1. Assemble the recovery record: the requirements and objectives, the backup schedule and catalog, both rehearsal results with the measured times, the decision tree, the DHCP and SQL restore notes, and a named owner with a scheduled next rehearsal that rotates through the set — an SBD-FS01 folder restored with its permissions, an SBD-SQL01 database with its logins, SBD-DHCP01 with its credential step — recovery is a practice with a calendar, not a binder. Keep its Tier 0 half in C:\Evidence\recovery on AdminJumpBox01 and its Tier 1 half on SBD-ADMIN-SRV01, each sealed with a manifest.
2. Take the closing snapshot: current Security Content Automation Protocol (SCAP) results for every system on the course 1.5 cadence, each tier from its own seat — Tier 0 from AdminJumpBox01, the Windows and Linux member servers from SBD-ADMIN-SRV01, the fleet from SBD-ADMIN-PC01 — with the recorded benchmark versions, triaged as always: the dated 'as-authorized' picture path 2 will reference. Credentialed vulnerability scanning joins it when course 1.17 (coming soon) arrives.
3. Reconcile the three living records one final time — the inventory on SBD-MGMT01 against the directory, DHCP leases, and WSUS rings (the course 1.15 method), the register against observed behavior, and the exception record against open findings — so the authorization path inherits agreement, not archaeology.
4. Write the Build path's closing note: what exists, where every record lives, and what is intentionally open — then start course 2.1, where this network stops being a project and becomes a system seeking authorization.

Check your result

```
# On AdminJumpBox01 (Tier 0), then the same on SBD-ADMIN-SRV01 (Tier 1): seal each half of
the recovery record
Get-ChildItem C:\Evidence -Directory | Select-Object Name
Get-ChildItem C:\Evidence\recovery -Recurse -File | Get-FileHash -Algorithm SHA256 |
Export-Csv C:\Evidence\recovery-manifest.csv -NoTypeInformation
Get-FileHash -Algorithm SHA256 C:\Evidence\recovery-manifest.csv
```

WHAT SUCCESS LOOKS LIKE The recovery record is complete with an owner and next-rehearsal date and sealed on both seats, the closing SCAP snapshot is filed with triage, and all three living records reconcile cleanly into the handoff note.

Your record

Evidence to keep: Finish the checklist: the recovery record and its two manifests, the closing SCAP snapshot, the three reconciliations, and the Build path's closing note.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

What makes recovery 'a practice with a calendar'?

> **REMEMBER** Recovery has an owner, a date, and a measured track record. The closing snapshot freezes the as-authorized picture. The Body of Evidence exists — path 2 assembles, it doesn't excavate.

Microsoft · Active Directory forest recovery guide

NIST · SP 800-53 Rev. 5 security controls (CP family)

Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

Lesson completion

LESSON	EVIDENCE FILED	DATE
01 Decide what recovery must prove	☐ yes	
02 Back up Tier 0: directory, CA, and policy	☐ yes	
03 Back up the member servers and records	☐ yes	
04 Carry the copies to the offline cabinet	☐ yes	
05 Rehearse a real restore, in isolation	☐ yes	
06 File the recovery record and close the path	☐ yes	

Open items carried forward

Next in the learning path

That completes the Build the Network path. Continue with course 2.1 to carry this network — and its evidence — through the Risk Management Framework to an authorization decision.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

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